Xiyu Ke

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Research Interests

Robotic manipulation and robot learning, with an emphasis on contact-aware planning and learned world models for predicting physical interaction outcomes.

Education

Rutgers University — M.S. in Electrical and Computer Engineering *Sep. 2025 – May 2027 (expected)*
GPA: **3.917/4.00**

Dalian Maritime University — B.S. in Intelligent Science and Technology *Sep. 2021 – Jun. 2025*
Weighted Average: 85.2/100 (Top 20%)

Publications

[1] **Xiyu Ke**, Litao Liu, Pengfei Yi, Yifan Han, Junshan Huang, Sixu Lin, and Jingjin Yu. "**CLEAR: Contact-Aware Localization and Exclusion-Aware Refinement**." *Submitted to ICRA 2027*. (First author)

[2] Junshan Huang, **Xiyu Ke**, Jintong Li, Litao Liu, Duo Zhang, and Jingjin Yu. "**OrganizeIt: Compiling Generated Images into Tidy Tabletop Arrangements**." *Submitted to AAAI 2027*. (Second author)

[3] Qiuyu Wang, Na Zhao, Chaojun Qin, **Xiyu Ke**, Yudong Luo, and Yantao Shen. "**Time-Optimal Trajectory Planning and Model Predictive Control of Morphing Quadrotors**." *ICRA 2026 (Accepted)*. (Fourth author)

[4] Litao Liu, Khushwanth Sri Sai Vudhya, Tanishq Annavaram, Pengfei Yi, Junshan Huang, Yifan Han, **Xiyu Ke**, Sixu Lin, and Jingjin Yu. "**Pushing Clutter via Attention-Based Multi-Object Interaction Dynamics**." *Submitted to ICRA 2027*.

[5] Litao Liu, Pengfei Yi, Enze Yuan, Zilin Yuan, Haoran Du, Ishaan Myrie, Yifan Han, Wenbo Yu, **Xiyu Ke**, and Jingjin Yu. "**DesignAgent: Efficient and Adaptive Search Framework for Robot Morphology Design**." *Submitted to AAAI 2027*.

Research Experience

CLEAR: Language-Guided Contact-Constrained Grasping *Jul. 2026 – Present*
Algorithmic Robotics and Control Lab, Rutgers University | *First author | Submitted to ICRA 2027*

- Developed a language-guided system for grasping user-specified object parts while avoiding forbidden-contact regions; deployed on an AgileX Nero arm with wrist-mounted RealSense D405 RGB-D sensing.
- Integrated reachable close-up RGB-D acquisition and automatic positive/negative point prompts for SAM3 to refine fine-grained object-part segmentation into target and forbidden contact maps.
- Implemented multi-start cross-entropy method (CEM) grasp refinement with target-contact rewards, forbidden-region penalties, and hard image-space/3D contact exclusions plus motion-feasibility checks.
- Evaluated 16 objects with 160 real-robot trials per method: **86.25% task success** versus **68.13%** for the strongest baseline; success required lifting, specified-part contact, and no forbidden contact.

OrganizeIt: Robotic Execution for Tabletop Organization *Jun. 2026 – Aug. 2026*
Algorithmic Robotics and Control Lab, Rutgers University | *Second author | Submitted to AAAI 2027*

- Developed AnyGrasp-based real-robot grasping and pick-and-place modules on a UR5e with wrist-mounted RGB-D sensing for tabletop rearrangement.
- Designed coordinated arm–gripper motion and object-specific gripper pre-shaping to pick up thin objects lying flush with the table.
- Implemented force-triggered recovery with re-perception and grasp retries after tipping, unexpected descent contact, or missed grasps; evaluated across four tabletop domains.

microMVP: Vision-Guided Multi-Vehicle Research Platform *Oct. 2025 – Mar. 2026*
Algorithmic Robotics and Control Lab, Rutgers University

- Built modular micro-vehicles using custom PCBs, embedded motor control, and 3D-printed chassis; developed Wi-Fi control and ChArUco-based localization.
- Implemented vision-guided multi-vehicle trajectory execution for circular, figure-eight, and user-drawn paths, demonstrating coordinated physical fleet motion.

Technical Skills

Programming & Learning: Python, C/C++, MATLAB, PyTorch, OpenCV; RGB-D perception, point-cloud processing, segmentation.

Robotics & Prototyping: ROS, CoppeliaSim (V-REP), grasp planning, SolidWorks, PCB design, 3D printing.